

Automated Student–Lab Allocation using Maximum Bipartite Matching

1. Project Overview

The **Automated Student–Lab Allocation using Maximum Bipartite Matching** project is designed to automate the allocation of students to available laboratory seats in an efficient and organized manner.

The system models students and individual lab seats as two sets of vertices in a **bipartite graph**. An edge represents a possible assignment between a student and a lab seat. The **Hopcroft–Karp algorithm** is used to find a maximum matching, ensuring that the maximum possible number of students are allocated while each student receives at most one lab seat.

The project also includes a separate **seating optimization module** that determines suitable physical seat positions within each allocated laboratory. The seating arrangement considers spacing between students and attempts to distribute students from the same branch across different areas of the laboratory.

2. Problem Statement

Manual student-to-lab allocation can become difficult when the number of students and available laboratory seats increases. It may result in uneven allocation, inefficient use of available seats, and difficulty in generating clear seating arrangements.

The proposed system automates this process by representing students and available lab seats using a bipartite graph and applying a maximum bipartite matching algorithm to obtain an efficient allocation.

3. Objectives

The main objectives of the project are:

- To automate student-to-lab allocation.
- To implement **Maximum Bipartite Matching** using the Hopcroft–Karp algorithm.

- To maximize the number of successfully allocated students.
- To ensure that each student is assigned to at most one laboratory seat.
- To consider laboratory capacity during allocation.
- To use branch information as a soft distribution factor.
- To optimize physical seating positions after lab allocation.
- To maintain reasonable spacing between students.
- To provide clear lab-wise seating visualization.
- To generate allocation reports and statistics.
- To provide a user-friendly interface for managing the allocation process.

4. Proposed Solution

The system accepts student and laboratory data as input files. Student records contain information such as roll number, name, and branch, while laboratory records contain laboratory names and available seating capacity.

The allocation process consists of two major stages:

Stage 1 – Lab Allocation

Students are represented on one side of a bipartite graph and individual laboratory seats on the other side.

The **Hopcroft–Karp algorithm** is applied to obtain a maximum matching between students and available seats. This maximizes the number of students that can be assigned without assigning more than one seat to the same student.

Branch-aware ordering is used as a soft distribution mechanism so that students from different branches can be distributed across laboratories where possible.

Stage 2 – Seating Optimization

After determining the laboratory for each student, a separate seating optimizer assigns physical seat numbers within that laboratory.

The physical seating layout remains fixed, while the optimizer attempts to:

- Spread students across available seats.
- Maintain spacing between students.
- Separate students from the same branch where possible.

- Adapt the seating arrangement according to the number of students.
- Keep all available seats visible in the seating plan.

The seating optimization is performed **after** maximum matching, so seating preferences do not reduce the number of students allocated.

5. Algorithm Used

Hopcroft–Karp Algorithm

Hopcroft–Karp is a maximum bipartite matching algorithm used to find the largest possible matching between two sets of vertices.

In this project:

Students $\leftrightarrow$ Laboratory Seats

Each student can be matched with at most one laboratory seat, and each seat can be assigned to at most one student.

The algorithm works by finding augmenting paths in phases:

1. Construct the bipartite graph.
2. Use BFS to identify the levels of vertices.
3. Use DFS to find augmenting paths.
4. Update the matching.
5. Repeat until no further augmenting path exists.

The resulting matching gives the maximum number of student-seat assignments supported by the available capacity and graph.

6. Seating Optimization

The project uses a separate **Seating Optimizer** after lab allocation.

Instead of simply assigning seats sequentially:

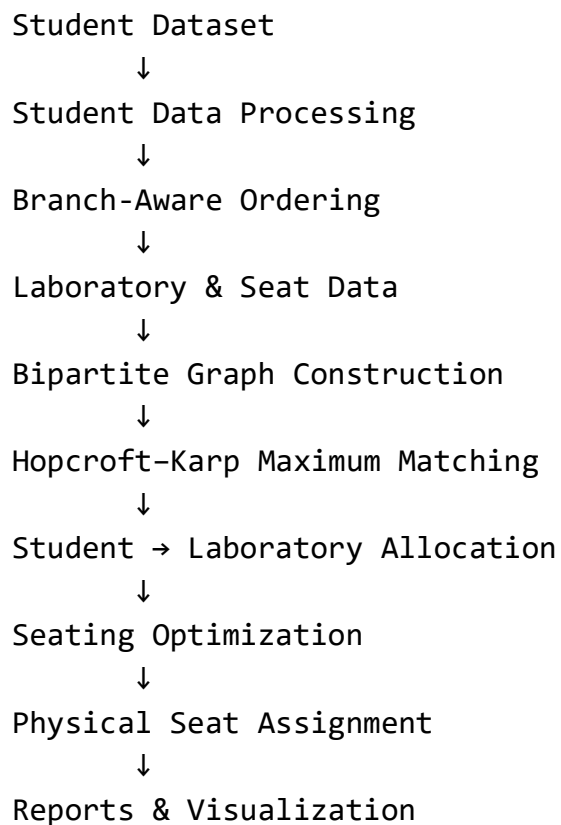
1 2 3 4 5
6 7 8 9 10
11 12 13 14 15

the optimizer attempts to distribute students across the physical grid to maintain better spacing.

For example, if only a few students are assigned to a laboratory, the system can place them at separated physical positions instead of filling the first consecutive seats.

This provides a more practical seating arrangement while preserving the maximum matching result.

7. System Workflow



8. System Features

The application provides the following features:

- Student dataset upload
- Laboratory dataset upload
- Automated allocation
- Maximum matching
- Branch-wise analysis
- Student search
- Lab-wise allocation information
- Reallocation
- Allocation statistics
- Physical seating plan
- Available-seat indication
- Seating visualization
- Allocation reports

9. Technology Stack

Backend

- **Java**
- **Maven**
- REST API
- Hopcroft–Karp Algorithm

Frontend

- **HTML5**
- **CSS3**
- **JavaScript**

Data

- CSV / Excel-based input data
- Student and laboratory datasets

10. Expected Output

The system produces a complete allocation containing:

- Student roll number
- Student name
- Branch
- Allocated laboratory
- Physical seat number
- Seating position
- Allocation statistics

The application also provides visual lab seating plans in which assigned seats are displayed along with unassigned seats marked as available.

11. Advantages

- Reduces manual allocation effort.
- Efficiently utilizes available laboratory capacity.
- Maximizes student allocation through maximum bipartite matching.
- Prevents duplicate student assignments.
- Provides organized laboratory-wise allocation.
- Produces optimized physical seating arrangements.
- Makes allocation results easier to understand through visualization.
- Can handle larger student and laboratory datasets compared with manual allocation.

12. Conclusion

The **Automated Student–Lab Allocation using Maximum Bipartite Matching** project provides an algorithm-based solution for automated laboratory allocation. By using the **Hopcroft–Karp algorithm**, the system efficiently obtains a maximum matching between students and available laboratory seats.

A separate seating optimization stage improves the physical arrangement of allocated students by considering spacing and branch distribution. The combination of algorithmic allocation, seating optimization, reporting, and visualization provides a structured solution for student laboratory management.